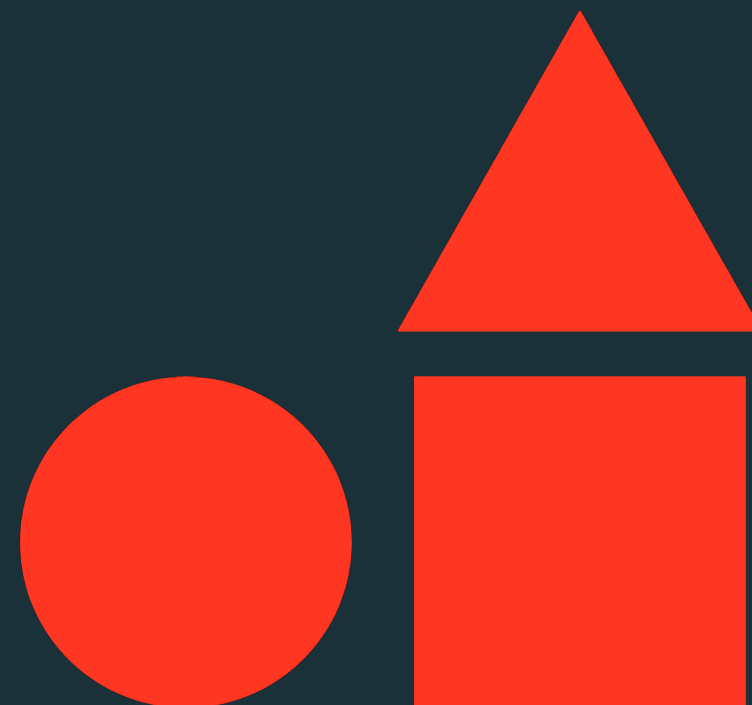


Northbay Energy Gauge Reader

AI visual inspection for
robot rounds

Starting with every pressure gauge | October 2026



Where the value comes from

Illustrative site: 500 gauges, photographed twice a day

~4,900 hours a year

- 365,000 readings a year, each transcribed or re-checked by hand: ~6,100 hours at 1 minute each¹
- With AI, people check only the uncertain ones (target: 1 in 5)
- About 3 people's time, freed for maintenance

~\$15k a year AI cost

- 365,000 photos × ~\$0.04 per photo at list price²
- vs. ~\$180k a year at a \$0.50 per-image vendor fee¹
- No model to train, host or maintain

Earlier warnings

- Excursions flagged the same shift, not at the next manual round
- Unplanned downtime costs a large plant ~\$0.8M an hour³
- Avoiding a single hour is worth ~50× the annual AI cost

¹ Assumptions to replace with Northbay's numbers: 1 min per reading, \$0.50 per image vendor fee. ² Measured tokens per photo × GPT-5.5 list price. ³ Siemens, The True Cost of Downtime 2024.



Value for both sides of the business

Built for the executive sponsor and the domain owner

Executive sponsor — VP Operations / HSE

- Robot rounds keep ~2,200 operator hours a year out of process areas (2 rounds a day × ~3 hours)
- Vendor dependency removed: ~\$15k a year on the platform vs. ~\$180k in per-image fees
- Unexpectedly high pressures surfaced the same shift, not at the next manual round

Domain owner — Reliability / Maintenance lead

- One review queue, sorted by risk: low confidence, unreadable, unexpectedly high
- Every correction audited (who, when, why) and fed back to the lakehouse
- Ask in plain English: “Are any pressure readings unexpectedly high?”



Why Northbay needs this

Robots photograph every gauge — trusted readings are the bottleneck

Analog gauges are here to stay

- The vast majority of plant gauges are still analog¹
- Swapping each for a transmitter means hot work, cabling and hazardous-area certification
- So the reading still has to come from a person, or a photo

A photo is not a reading

- Robots already capture photos at scale: 120,000+ inspections a month for one vendor alone³
- Glare, dirt, rain and steep angles defeat simple vision models¹
- Third-party readings still get re-checked by hand, outside the data platform

Late readings cost money

- An out-of-range pressure found at the next manual round is a missed early warning
- Unplanned downtime costs the world's 500 largest companies ~\$1.4tn a year, 11% of revenue²

¹ Reitsma et al, "Under pressure: learning-based analog gauge reading in the wild", ETH Zurich, ICRA 2024. ² Siemens, The True Cost of Downtime 2024. ³ ANYbotics, anybotics.com (vendor-reported).



What we hear from energy operators

Patterns from robot inspection programmes, and our answer to each

What operators tell us	How Northbay's Gauge Reader answers it
<i>"Accuracy comes first — a wrong reading is worse than a late one."</i>	Every reading carries a confidence score; low-confidence, unreadable and unexpectedly high readings always go to a person
<i>"We pay a vendor per image, and still check its readings by hand."</i>	Readings run inside the platform at a known cost per reading (~\$0.04), with every correction audited
<i>"Photos are taken in fog, rain and glare, on dirty dials."</i>	The model reports image issues (glare, blur, angle, dirt) so reviewers see why a reading was flagged
<i>"We want to own the capability, not rent a black box."</i>	A governed foundation model in Unity Catalog: no training data, swap models through Unity AI Gateway
<i>"Next: spot rust and leaks, and trends across rounds."</i>	The same pipeline extends to condition findings and per-gauge trends, feeding maintenance work orders



One platform for visual inspection

Gauge readings are the first use case — not the last

Beyond gauges

- The same photos also show leaks, corrosion and rust, breakages, missing guards and valve positions
- A new inspection type is a new prompt and output fields, not a new project
- Same review queue, same governance, same Genie

Open, not a black box

- Foundation vision models instead of hand-built computer vision that needs tuning per dial, angle and light
- No third-party vendor or per-image fee; prompts, data and results stay in your lakehouse
- Swap models through Unity AI Gateway

Specialise when it pays

- The model is an example: start with a foundation model, no training needed
- Every reviewer correction becomes a labelled example
- Fine-tune specialist models (gauge reading, rust detection) and evaluate them before switching

Any cloud, any robot

- Runs on Databricks on AWS, Azure and Google Cloud, deployed as one bundle¹
- Works with photos from any robot, drone or handheld camera
- No lock-in to a robot or vision vendor

¹ Lakebase is in Beta on Google Cloud; where Lakehouse Sync (Lakebase → Delta) is not yet available, a scheduled MERGE returns review decisions to Delta (DECISIONS D4).



Beyond classic vision and vendors

Foundation models on an open platform change the economics

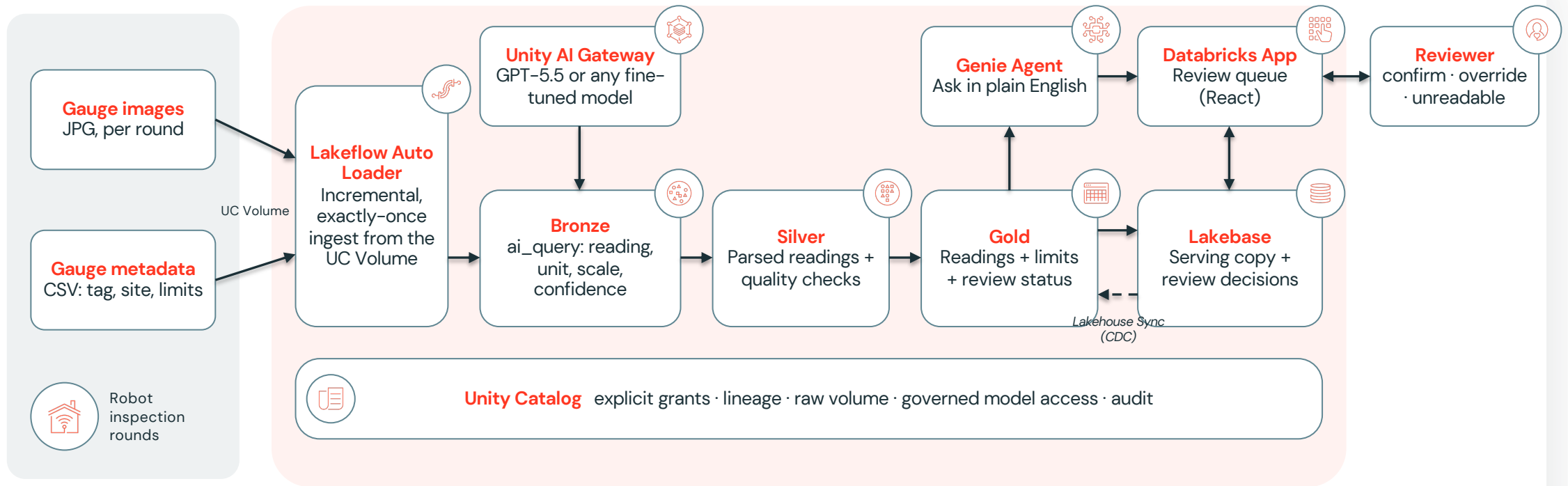
	Bespoke computer vision	Third-party vendor	Foundation models on Databricks
New inspection type	New detection, segmentation and OCR models to build	Wait for the vendor's roadmap	New prompt and output fields; test it the same day
Labelled data to start	Labelled images for every gauge type and defect	None, but it's the vendor's model	None — reviewer corrections build a labelled set as you go
Hand-tuning	Per dial design, angle, glare and lighting	Opaque; results still re-checked by hand	Confidence score routes uncertain readings to a person
Cost model	GPU training plus a team to maintain it	Fee per image	Pay per token: ~\$0.04 per gauge reading²
Data and governance	A separate stack to secure	Results arrive outside your platform	Unity Catalog: grants, lineage, audit
Improves over time	Only when someone retrains it	On the vendor's terms	Swap or fine-tune models through Unity AI Gateway

² Build run: GPT-5.5 via Unity AI Gateway at list price (evidence/07_gateway.md). Bespoke pipeline steps as described in Reitsma et al., ETH Zurich, ICRA 2024.



Architecture

One governed journey: robot photo → trusted reading → answer



Human in the loop confidence < 70%, unreadable or unexpectedly high → a reviewer; decisions appended, never overwritten

Unity Catalog analysts see gold only; the app reads photos read-only; Genie runs as the signed-in user

Unity AI Gateway every model call attributed and costed (~\$0.04 per reading); swap models without pipeline changes



Decisions and trade-offs

Why this approach and not another

- Foundation vision model via ai_query, not a custom CNN: no labelled data or training; confidence-based review covers the gaps
- Model called inside the streaming table: each image read and billed once; reviewed readings never change on refresh
- Lakebase for the review step: low-latency transactional writes; decisions return to Delta via Lakehouse Sync
- Snapshot sync for serving: gold is a materialized view; at ~120 images/day a snapshot takes seconds
- “Unexpectedly high” = above the gauge's operating limit (default 75% of full scale), so Genie never invents thresholds
- GPT-5.5 chosen after testing 15 vision models: best accuracy among batch-capable models, with a confidence score that separates right from wrong



Roadmap and pilot

Each phase reuses the pipeline, review app, governance and Genie

- 1 · Gauge readings: pilot at one site; unexpectedly high readings raise maintenance work orders
 - 2 · Condition findings: leaks, rust and breakages from the same photos, in the same review queue
 - 3 · Specialist models: fine-tune on reviewer-labelled photos; evaluate before switching
 - Pilot set-up: connect the robots' image export, load gauge tags and limits from the asset register, promote dev → prod with bundles
- Pilot success criteria
 - One site, 8 weeks, ~500 gauges
 - At most 1 in 5 readings sent to review
 - $\geq 95\%$ of auto-accepted readings within $\pm 2\%$ of full scale (weekly spot check)
 - ~4,900 hours a year of reading and re-checking removed (run rate)
 - Every reading traceable from photo to answer



**Every photo read, every finding
governed**

